Stony Brook Institute at Anhui University

安徽大学纽约石溪学院

Chapter 1 Random events and its probability

Section 3 Probability of Random Events

- Definition and property of frequency
- Statistical definition of probability
- Axioms of probability
- Classical probability
- Geometric probability

Definition and property of frequency

Definition 1.3.1: The frequency of an event A is defined by

$$f_n(A) = \frac{n_A}{n}$$

where n_A is the number of times that A occurs, n is the total number of random trials.

The frequency of a event A has the following properties:

1. (**Non-negativity**) $0 \leq f_n(A) \leq 1$
2. (**Norming**) $f_n(\Omega) = 1$, $f_n(\Phi) = 0$
3. (**Countable Additivity**) If A_1; A_2; $\cdots$; A_m are mutually exclusive events, then we have

$$f_n(A_1 \cup A_2 \cup \cdots \cup A_m) = f_n(A_1) + f_n(A_2) + \cdots + f_n(A_m)$$

Definition and property of frequency

Example 1.3.1: Tossing a fair coin 5, 50 and 500 times, recording the number of frontage on the top.

试验序号	$n=5$		$n=50$		$n=500$	
	n_H	f	n_H	f	n_H	f
1	2	0.4	22	0.44	251	0.502
2	3	0.6	25	0.50	249	0.498
3	1	0.2	21	0.42	256	0.512
4	5	1.0	25	0.50	247	0.494
5	1	0.2	24	0.48	251	0.502
6	2	0.4	18	0.36	262	0.524
7	4	0.8	27	0.54	258	0.516

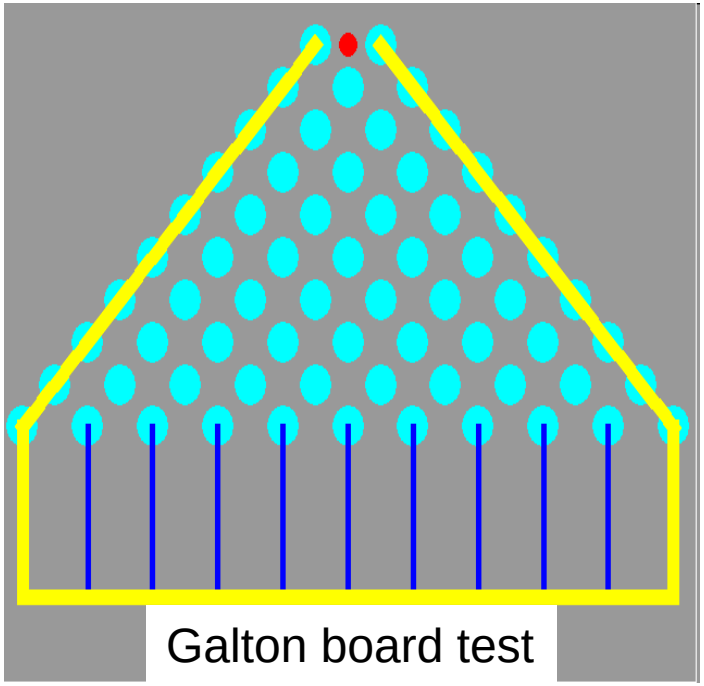
Definition and property of frequency

n	n_H	f
2048	1061	0.5181
4040	2048	0.5069
12000	6019	0.5016
24000	12012	0.5005

Note:

- For same *n*, frequency may be different.
- When the number of coin toss *n* is small, the random fluctuation of frequency *f* is large, but with the increase of *n*, the frequency *f* shows stability. That is, when *n* increases gradually, the frequency *f* is always around 0.5 and gradually stabilizes at 0.5. This stable value represent the chance of this event, i.e. probability.

Definition and property of frequency



- The small balls fall freely from the upper board.
- When the ball falls, it hits a nail and has an equal chance of falling from both sides.
- It is the same when it hits the next row of nails.
- Finally, it falls into a grid in the bottom plate.

Section 3 Probability of Random Events

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Statistical definition of probability

Definition 1.3.2: In random trials, when the number of trials increase, the frequency of event A converges to a certain constant p, $0 \leq p \leq 1$, we define p as the probability of event A, marked as $P(A)=p$.

The probability of a event A has the following properties:

1. (**Non-negativity**) $0 \leq P(A) \leq 1$
2. (**Norming**) $P(\Omega)= 1$, $P(\Phi)= 0$
3. (**Countable Additivity**) If A_1; A_2; $\cdots$; A_m are finite mutually exclusive events, then we have

$$P(A_1 + A_2 + \cdots + A_m) = P(A_1) + P(A_2) + \cdots + P(A_m)$$

Note: The statistical definition of probability intuitively describes the probability of the occurrence of an event and reflects the essence of probability, but it also has some shortcomings, that is, it cannot calculate the probability of an event according to this definition.

Definition and property of frequency

Shortcoming of the frequency interpretation

- "repeated a large number of times under similar conditions" are too vague to serve as the basis for a scientific definition of probability.
- it applies only to a problem in which there can be, at least in principle, a large number of similar repetitions of a certain process. Many important problems are not of this type.

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Axioms of probability

Introduction

In 1933, The Soviet mathematician Kolmogorov put forward the axiomatic structure of probability theory and gave a strict definition of probability, which made the rapid development of probability theory.

Definition 1.3.6: A probability on a sample space Ω is a function P : A(Ω) ∈ R that assigns numbers to events, where A(Ω) is the set of all the events of the experiment, and the map P(·) must satisfy the following three axioms

1. (**Non-negativity**) $0 \leq P(A) \leq 1$
2. (**Norming**) $P(\Omega) = 1$
3. (**Countable Additivity**) If A_1; A_2; · · ·; A_m are mutually exclusive events, then we have

$$P(A_1 + A_2 + \cdots + A_m) = P(A_1) + P(A_2) + \cdots + P(A_m)$$

Axioms of probability

(1) $P(\varnothing) = 0.$

$$A_n = \varnothing \ (n = 1, 2, \cdots),$$

$$\Rightarrow \bigcup_{n=1}^{\infty} A_n = \varnothing, \qquad A_i A_j = \varnothing, \quad i \neq j.$$

$$P(\varnothing) = P\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} P(A_n)$$

$$\left.\begin{aligned} &= \sum_{n=1}^{\infty} P(\varnothing) \\ &P(\varnothing) \geq 0 \end{aligned}\right\} \Rightarrow P(\varnothing) = 0.$$

Axioms of probability

(2) Suppose $A_1, A_2 \cdots A_n$ are disjoint events,

$$\textit{Let}\ \ A_{n+1} = A_{n+2} = \cdots = \varnothing,$$

$$\Rightarrow A_i A_j = \varnothing,\ \ i \neq j, \quad i, j = 1,2,\cdots.$$

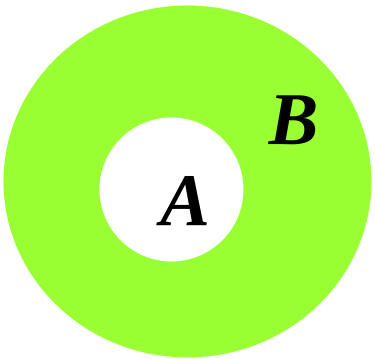
$$P(A_1 \cup A_2 \cup \cdots \cup A_n) = P(\bigcup_{k=1}^{\infty} A_k) = \sum_{k=1}^{\infty} P(A_k) = \sum_{k=1}^{n} P(A_k) + 0$$

$$= P(A_1) + P(A_2) + \cdots + P(A_n).$$

Axioms of probability

(3) Let A; B be two events, and A ⊂ B,

$$A \subset B,$$



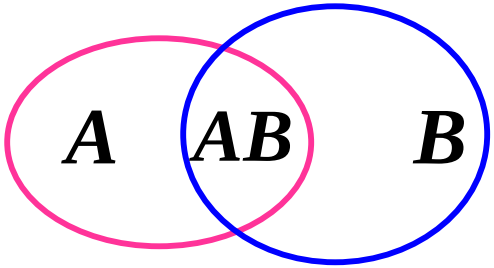
$$\Rightarrow \boxed{P(B-A) = P(B) - P(A).}$$

$$P(B-A) \geq 0,$$

$$\Rightarrow P(A) \leq P(B).$$

Axioms of probability

(4) Let A; B be two events,



$$A \cup B = A + (B - AB),$$

$$\Rightarrow P(A \cup B) = P(A) + P(B) - P(AB).$$

$$P(A_1 \cup A_2 \cup A_3)$$
$$= P(A_1) + P(A_2) + P(A_3) - P(A_1A_2) - P(A_2A_3)$$
$$- P(A_1A_3) + P(A_1A_2A_3).$$

Axioms of probability

Probability of a union of n events:

$$P(A_1 \cup A_2 \cup \cdots \cup A_n) = \sum_{i=1}^{n} P(A_i) - \sum_{1 \le i < j \le n} P(A_i A_j)$$

$$+ \sum_{1 \le i < j < k \le n} P(A_i A_j A_k) + \cdots + (-1)^{n-1} P(A_1 A_2 \cdots A_n).$$

Axioms of probability

Property of probability:

1.The probability of the impossible event is zero, P(Φ) = 0.

2. (**Probability is finitely additive**)Suppose A_1, A_2 $\cdots A_n$ are **disjoint events**, then

$$P\ (A_1\ \cup A_2\ \cup \cdots \cup\ An)\ =\ P\ (A_1) + P\ (A_2) + \cdots + P\ (An)$$

3. (**Monotonicity**) Let A; B be two events, and A ⊂ B, then

$P(B\ - A) = P(B) - P(A)$, furthermore we have P(A) ≤ P(B).

4. P(A) ≤ 1 for any event A.

5. (Probability of Complement event) $P(A^c) = 1 - P(A)$

6. (Probability of a union of events) P(A ∪ B) = P(A) + P(B) - P(AB)

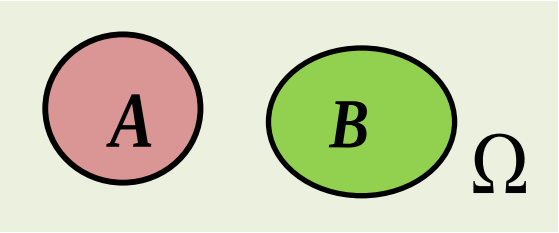
Examples

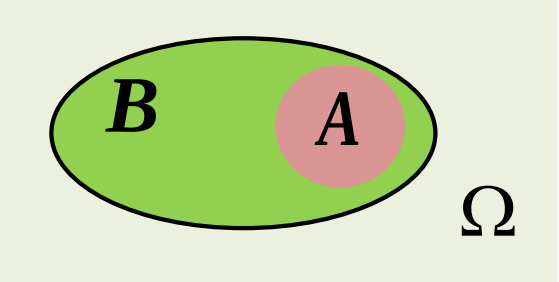
Examples 1.3.6: $P(A)=\frac{1}{3}, P(B)=\frac{1}{2}$, calculate $P(B\bar{A})$ when

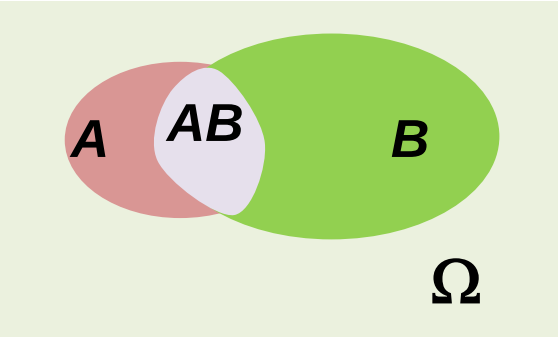
(1) A and B are disjoint

(2) A ⊂ B

(3) P(AB)=1/8







Answer:

(1) $\mathrm{P}(\mathrm{B}\bar{\mathrm{A}}) = \mathrm{P}(\mathrm{B}) = \frac{1}{2}$

(2) $\mathrm{P}(\mathrm{B}\bar{\mathrm{A}}) = \mathrm{P}(\mathrm{B}) - P(A) = \frac{1}{6}$

(3) $\mathrm{P}(\mathrm{B}\bar{\mathrm{A}}) = P(B) - P(AB) = \frac{3}{8}$

Summary: $\mathrm{P}(\mathrm{B}\bar{\mathrm{A}}) = P(B) - P(AB)$

P(AB) is different in above 3 situations.
What is the P(AB) in above 3 situations?

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Classical probability

Definition 1.3.3: If a randomized trial E has the following characteristics：

1. The sample space of the test contains only a limited number of sample points;
2. Each sample point is **equally likely** to appear.

Then the randomized experiment is called **classical model.**

Classical probability

Definition 1.3.4: Let the sample space of test E be composed of n sample points, A is any event of E and contains m sample points, then the probability of occurrence of event A is denoted as:

$$P(A) = \frac{m}{n}$$

We call it the classical interpretation of probability (classical probability).

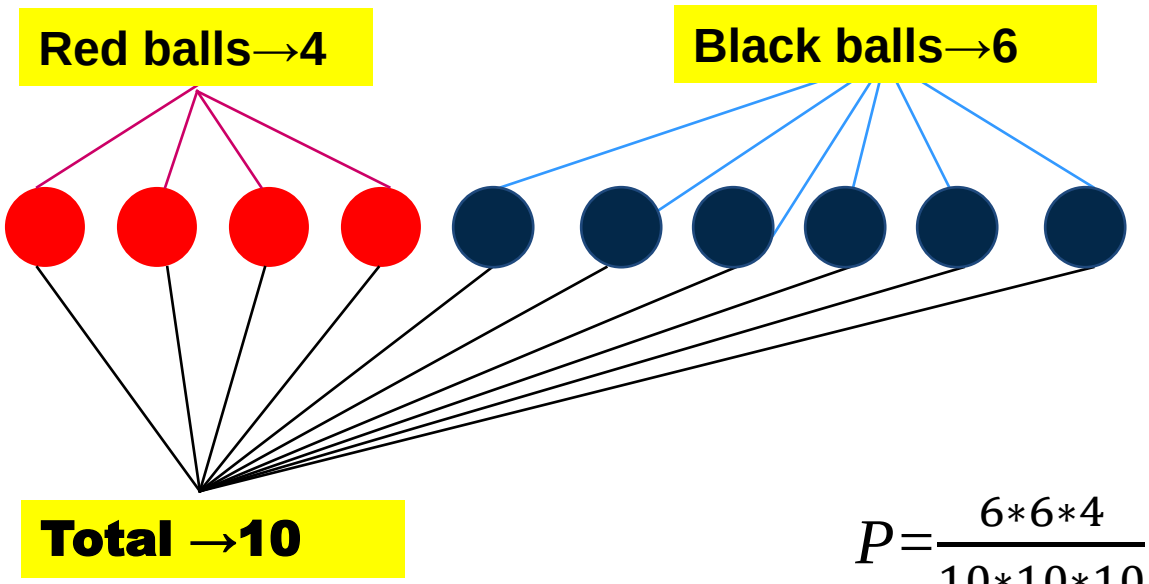
The classical probability has the following **properties**:

1. (**Non-negativity**) $0 \leq P(A) \leq 1$
2. (**Norming**) $P(\Omega)= 1$, $P(\Phi)= 0$
3. (**Countable Additivity**) If A_1; A_2; $\cdots$; A_m are finite mutually exclusive events, then we have

$$P(A_1 + A_2 + \cdots + A_m) = P(A_1) + P(A_2) + \cdots + P(A_m)$$

Examples

Example 1.3.1 (with return): There are 4 red balls and 6 black balls in the bag. Sampling the balls 3 times randomly, and calculate the probability of sampling the black ball in the first two times and the red ball in the third time?



$$P=\frac{6*6*4}{10*10*10}=0.144$$

Examples

Example 1.3.2 (without return): There are M white balls and N black balls in the bag. take m+n balls one by one, what is the probability of the taken balls' color are m white and n black?

Answer:

Let A={the color of taken balls are m white and n black}

The total number of sample space is $\binom{M+N}{m+n}$

The number of sample points of event A is $\binom{M}{m} \cdot \binom{N}{n}$

Then $P(A) = \frac{\binom{M}{m} \cdot \binom{N}{n}}{\binom{M+N}{m+n}}$

Examples

Example 1.3.3 (Counting/Combinatorial methods): Fifteen new students were randomly assigned equally to three classes, and three of them were boys. (1) What is the probability that each class will be assigned a boy?
(2) What is the probability that three boys are assigned to the same class?

Answer:

The number of outcomes when 15 new students are assigned equally to three classes: $\binom{15}{5}\binom{10}{5}\binom{5}{5}=\frac{15!}{5!5!5!}$.

(1) The number of outcomes when each class is assigned one boy: $\binom{3}{1}\binom{2}{1}\binom{1}{1}\binom{12}{4}\binom{8}{4}\binom{4}{4}=\frac{12!3!}{4!4!4!}$, then $P(1)=\frac{\frac{12!3!}{4!4!4!}}{\frac{15!}{5!5!5!}}=\frac{25}{91}$.

(2) Then number of outcomes when 3 boys in the same class: $\binom{12}{2}\binom{10}{5}\binom{5}{5}=\frac{12!}{2!5!5!}$, then $P(2)=\frac{\frac{12!}{2!5!5!}}{\frac{15!}{5!5!5!}}=\frac{6}{91}$.

Examples

Examples 1.3.7 (Combinatorial methods) : There are 5 pairs of shoes, we take 4 shoes randomly, what is the probability of at least 1 pair of shoes among the 4 taken shoes?

Answer:

Let A={at least 1 pair of shoes among the 4 taken shoes}

A_1={1 pair of shoes among the 4 taken shoes}

A_2={2 pair of shoes among the 4 taken shoes}

$A_1A_2 = \Phi$ and $A_1 + A_2 = A$

$$\mathrm{P(A)} = \mathrm{P}(A_1) + \mathrm{P}(A_2) = \frac{C_5^1[C_4^2 2^2]}{C_{10}^4} + \frac{C_5^2}{C_{10}^4} = \frac{13}{21}$$

Another way:

$\bar{A}$={no shoes among the 4 taken shoes are in pairs}

$$P(A) = 1 - P(\bar{A}) = 1 - \frac{C_5^4 2^4}{C_{10}^4} = \frac{13}{21}$$

Homework

Classwork 1: Six different numbers are arbitrarily extracted from the 10 numbers 0, 1, 2, 3,..., 9.

Question: the probability that these six numbers are arranged into six digits is even.

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Geometric probability

- The classical probability has the advantage of computability, but it also has obvious limitations, i.e. the number of sample points must be finite. If the number of sample points in sample space is infinite, then the classical probability theory doesn't apply.

- In order to extend a finite number of sample points to a infinite number of sample points, geometric probability theory is introduced to determine the probability.

Geometric probability

Definition 1.3.4: For a random trial, each sample point is equally likely to appear, and the number of sample points in sample space is infinite, but can be measured in terms of geometric methods, i.e. $0 < \mathrm{m}(\Omega) < \infty$. This type of random trial is called Geometric model probability.

Definition 1.3.5: When the sample space of a randomized trial is a certain area, and any sample point falls in the sub-area with the same measurement (length, area, volume) is equally likely, then the probability of event A can be defined as

$$P(A) = \frac{m(A)}{m(\Omega)}$$

where $m(A)$ is the measurement of event A, and $m(\Omega)$ is the measurements of sample space.

Geometric probability

The property of geometric probability

1. (**Non-negativity**) $0 \leq P(A) \leq 1$

2. (**Norming**) $P(\Omega)= 1$, $P(\Phi)= 0$

3. (**Countable Additivity**) If A_1; A_2; $\cdot\cdot\cdot$; A_m are mutually exclusive events, then we have

$$P(A_1 + A_2 + \cdots + A_m) = P(A_1) + P(A_2) + \cdots + P(A_m)$$

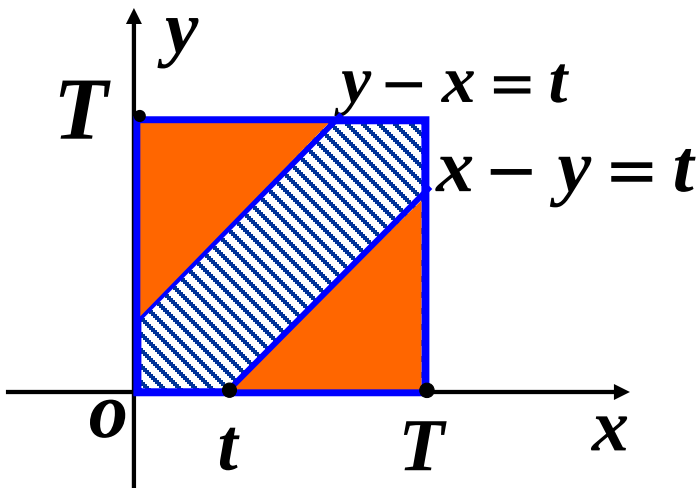
Geometric probability

Example 1.3.4: A and B make an appointment and are going to meet each other during the period 0~T, The first person to arrive will wait for the other one, and left after time t(t< T). Assuming that the arrival time during 0~T is equally likely, and arrival time of A and B are irrelevant. What is the probability A and B can meet each other.

Answer:

Let x and y be the arrival time for A and B. So 0<x<T, 0<y<T. When $|x-y|<t$, A and B can meet each other.

(x, y) in the shadow is A and B can meet each other.

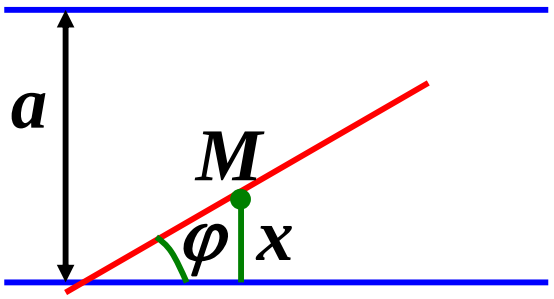


$$P=\frac{area\ of\ shadow}{area\ of\ square}$$

$$=\frac{T^2-(T-t)^2}{T^2}=1-(1-\frac{t}{T})^2$$

Geometric probability

Example 1.3.5: **Buffon needle trial**. In 1777, Buffon, a French scientist, proposed the problem of the throwing needle test. Some parallel lines are drawn on the plane equidistantly, the interval is a (>0). Now, a needle with length b(<a) is thrown on the plane. What is the probability of the intersection of the needle with any parallel line?



Answer:

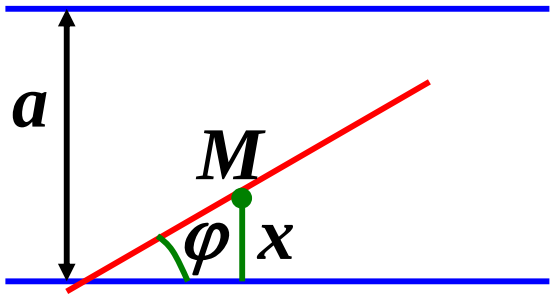
Set M is the middle point of the needle, x is the distance between M and the line nearest to M

φ is the angle between the needle and parallel lines.

(x, φ) determines the intersection between the needle and the nearest line, which is the intersection between the needle and the parallel lines actually.

Geometric probability

Example 1.3.5: **Buffon needle trial**.



Answer:

The intersection of the needle with any parallel line ↔ intersection of the needle with the nearest line.

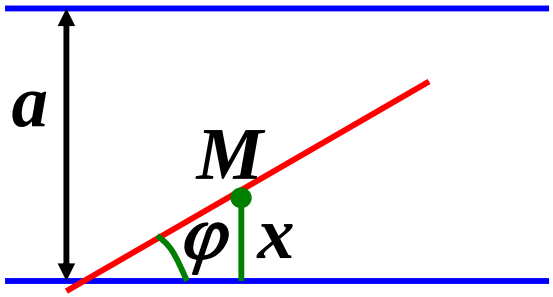
The sample space is $\Omega = \{(x, \varphi) | 0 \le x \le \frac{a}{2}, 0 \le \varphi \le \pi\}$

A={intersection of the needle with any parallel line}={intersection of the needle with the nearest line}

A ↔ $\{(x, \varphi) | 0 \le x \le \frac{b}{2}\sin(\varphi), 0 \le \varphi \le \pi\}$

Geometric probability

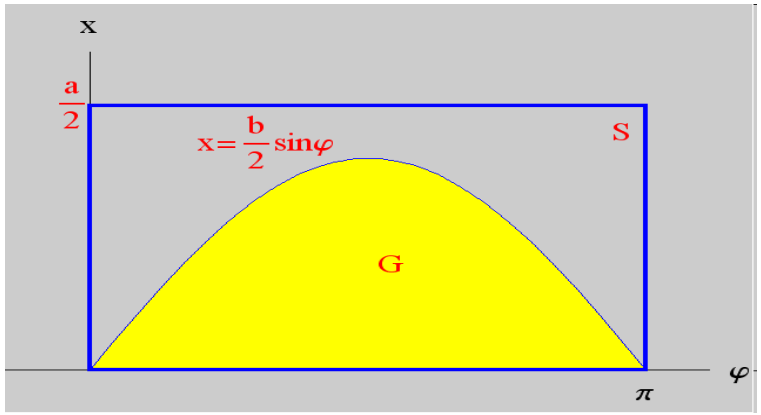
Example 1.3.5: Buffon needle trial.



Answer:

$$\{(x,\varphi)|0 \le x \le \frac{b}{2}\sin(\varphi), 0 \le \varphi \le \pi\}$$

Area G!



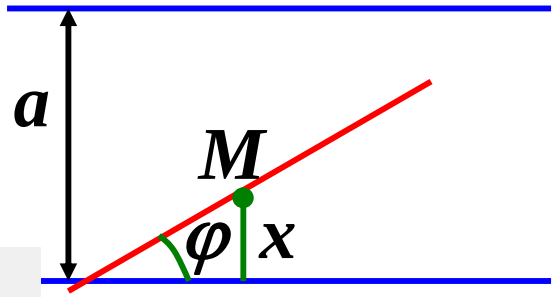
$$P(A) = \frac{m(G)}{m(\Omega)} = \frac{area\ of\ G}{area\ of \Omega}$$

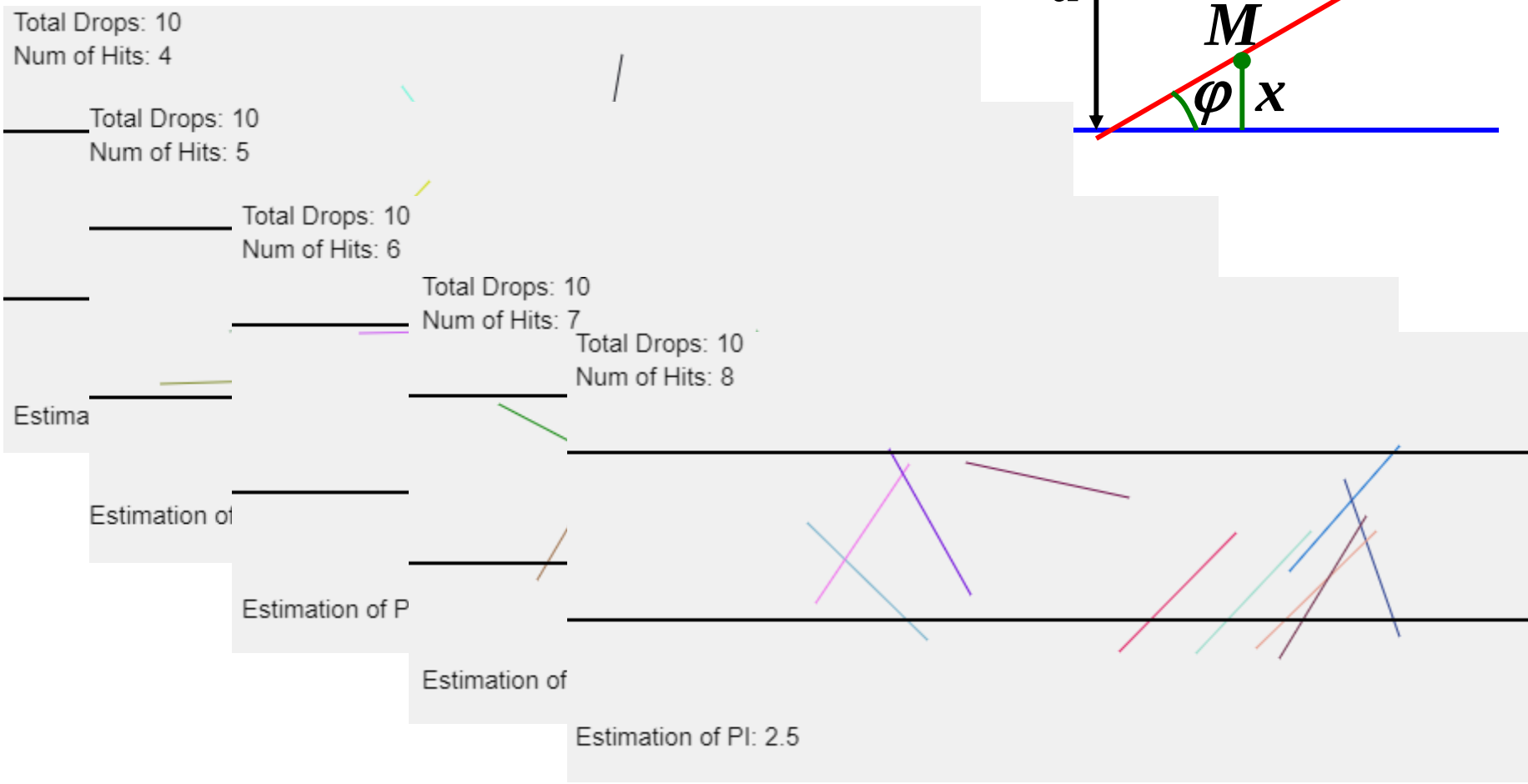
$$= \frac{\int_0^{\pi} \frac{b}{2}\sin(\varphi)\, d\varphi}{\frac{a}{2}\pi} = \frac{b}{\frac{a}{2}\pi} = \frac{2b}{a\pi}$$

N times Buffon needle trials, when N is very big, frequency $f(A) \approx P(A) = \frac{2b}{a\pi} \to \pi \approx \frac{2b}{af(A)}$ can be used to estimate π.

Geometric probability

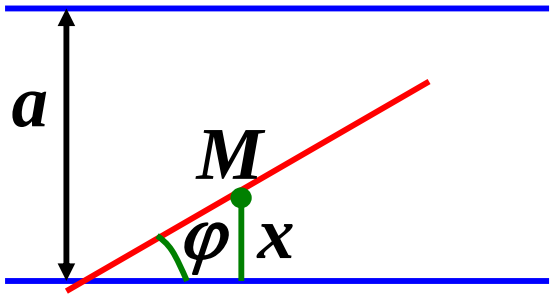
Example 1.3.5: Buffon needle trial.

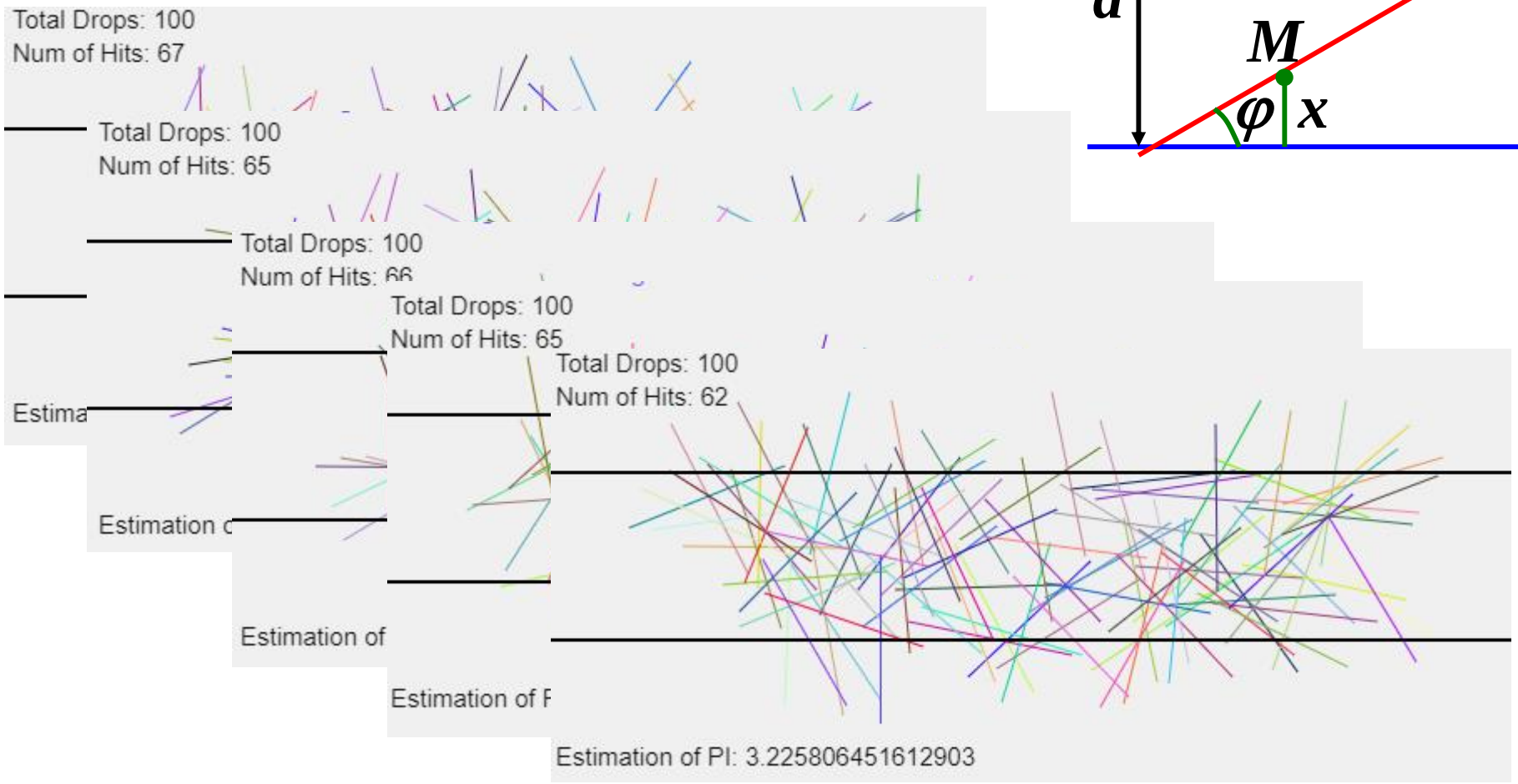




Geometric probability

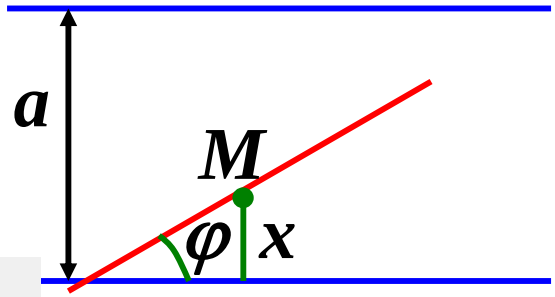
Example 1.3.5: Buffon needle trial.

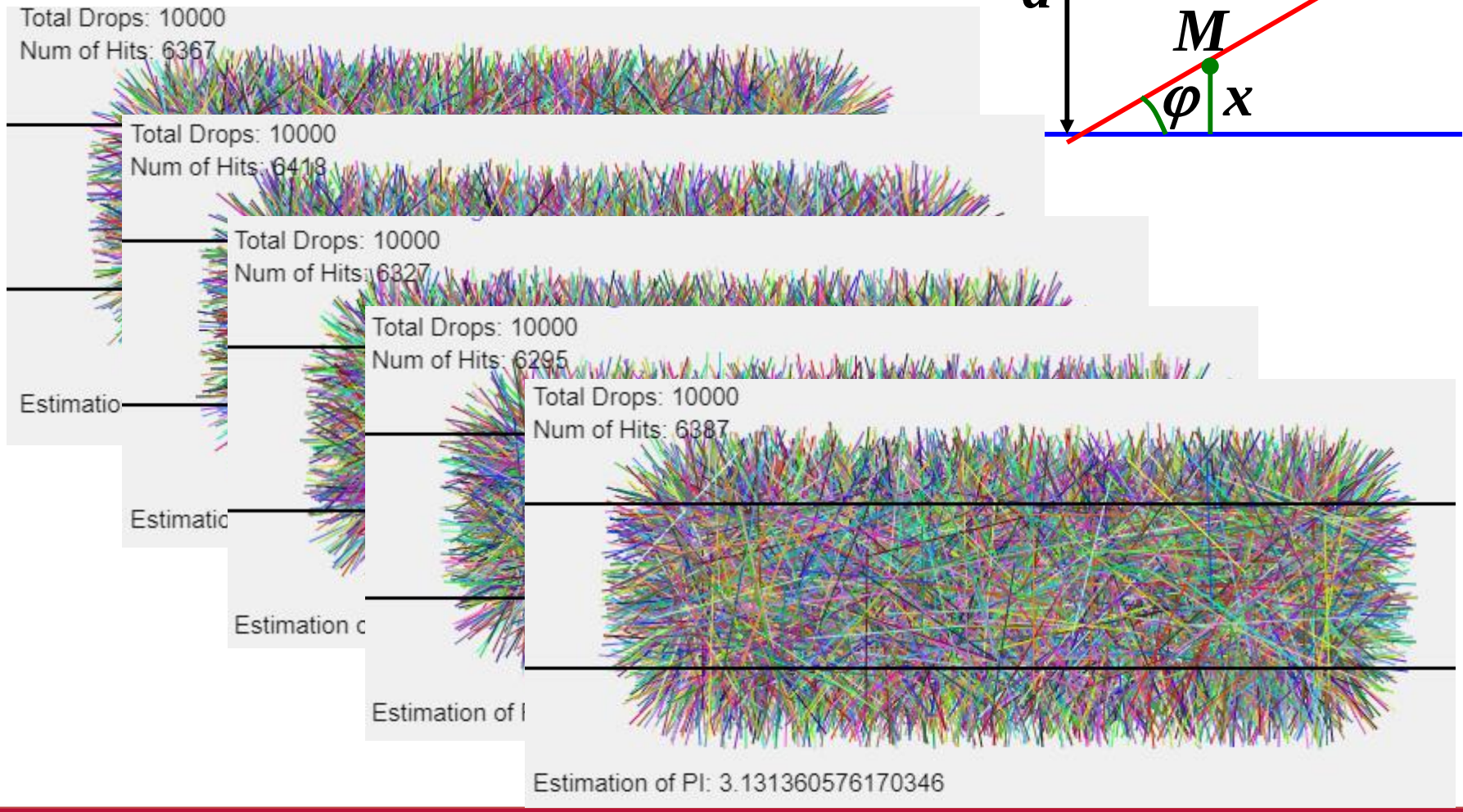




Geometric probability

Example 1.3.5: Buffon needle trial.





Homework:

Chapter 1.5(P21) :Q7 & Q11
Chapter 1.7(P32) : Q10
Chapter 1.8(P41) : Q13